

United International University

Department of Computer Science and Engineering

AI-Based Fire Fighting and Human Detection Rescue Robot

Using Raspberry Pi and Arduino

Course Title: Introduction to Robotics / Project
Department of Computer Science and Engineering

Team Members

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GitHub Repository:

<https://github.com/shafinSI/AI-Fire-Fighting-Human-Detection-Rescue-Robot>

Abstract

Fire accidents and emergency situations often put human lives at risk. Entering hazardous environments during a fire can be extremely dangerous for firefighters and rescue teams. To address this problem, we developed an AI-Based Fire Fighting and Human Detection Rescue Robot using Arduino Uno and Raspberry Pi.

The robot can be controlled remotely through Bluetooth communication. A flame sensor is used to detect fire, while an ultrasonic sensor detects nearby obstacles. When fire is detected, a relay module automatically activates a water pump to extinguish the fire. A servo motor directs the water nozzle toward the fire source. Raspberry Pi performs human detection using computer vision techniques and sends alerts to the robot controller.

The proposed system is low-cost, efficient, and suitable for industrial safety, rescue operations, and emergency response applications.

1. Introduction

Fire accidents frequently occur in homes, industries, warehouses, and commercial buildings. During rescue operations, firefighters often face life-threatening situations while entering affected areas.

Recent developments in robotics and artificial intelligence have made it possible to create intelligent systems capable of assisting rescue teams. A rescue robot can reduce human risk by entering dangerous locations and performing essential tasks such as fire detection, obstacle monitoring, and human identification.

This project presents an AI-Based Fire Fighting and Human Detection Rescue Robot that combines Arduino Uno and Raspberry Pi to provide intelligent monitoring and automated fire-fighting capabilities. The robot uses a tank-track chassis for mobility over various surfaces, making it suitable for real-world rescue environments.



Figure 1: Fully Assembled Rescue Robot with Tank Chassis

2. Problem Statement

During fire emergencies, rescue personnel are exposed to dangerous conditions while attempting to save lives and property. Manual fire-fighting operations are risky, time-consuming, and often inefficient in hazardous environments.

Most low-cost fire-fighting robots available today lack advanced features such as human detection and intelligent monitoring. Therefore, there is a need for a smart robotic system capable of detecting fire, locating humans, and assisting rescue operations safely and effectively.

3. Objectives

The main objectives of this project are:

- To develop a Bluetooth-controlled rescue robot with tank-track mobility.
- To detect fire using a flame sensor and respond automatically.
- To extinguish fire automatically using a relay-controlled water pump.
- To detect obstacles using an ultrasonic sensor.
- To detect humans using Raspberry Pi and AI-based computer vision (YOLO).
- To control water nozzle direction using a servo motor.
- To improve safety and efficiency during rescue operations.

4. Proposed System

The proposed system consists of two major units working together:

4.1 Arduino Unit

The Arduino Uno is responsible for:

- Bluetooth communication via HC-05 module
- Motor control using two BTS7960 / IBT-2 motor drivers
- Flame detection via flame sensor
- Obstacle detection via ultrasonic sensor
- Relay control for water pump activation
- Servo motor control for nozzle direction
- Buzzer alerts

4.2 Raspberry Pi Unit

The Raspberry Pi 4 is responsible for:

- Camera image capture and processing
- Real-time human detection using YOLO model
- AI-based computer vision analysis
- Serial communication with Arduino

The combined system allows intelligent monitoring and automated emergency response.

5. System Architecture / Block Diagram

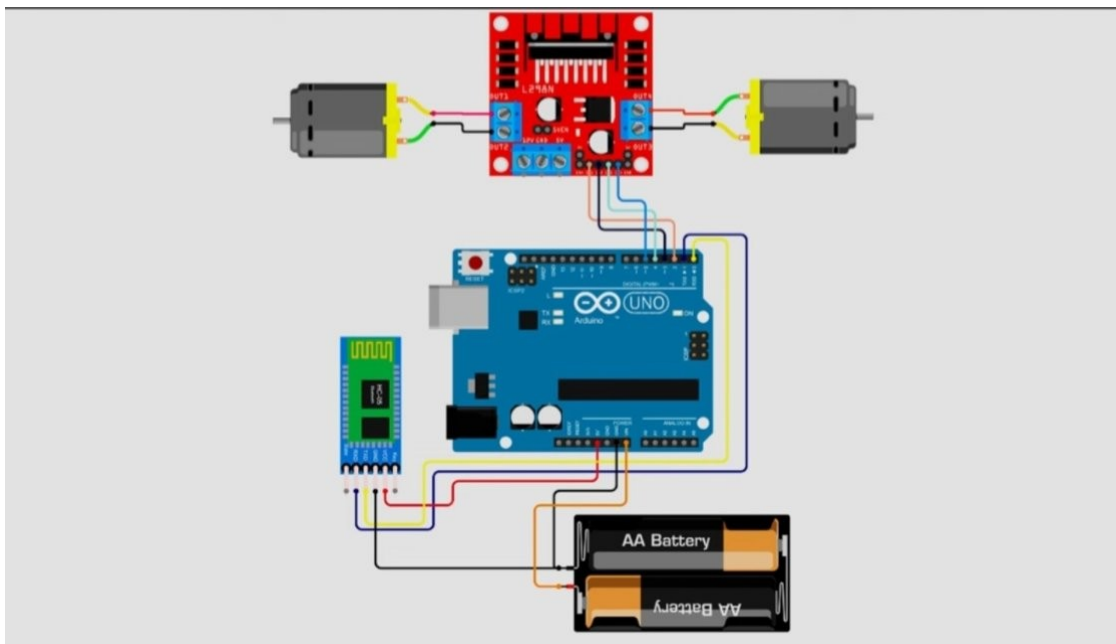


Figure 2: Circuit Connection Diagram (Arduino UNO + L298N + Bluetooth)

System Flow:

- Bluetooth App → Arduino Uno → Motor Drivers → DC Motors (Tank Movement)
- Flame Sensor → Arduino → Relay Module → Water Pump
- Ultrasonic Sensor → Arduino → Buzzer / Stop
- Raspberry Pi Camera → YOLO Detection → Serial → Arduino → Stop / Alert

6. Hardware Components

The following hardware components were used in the development of the rescue robot:

Component	Quantity
Arduino Uno R3	1
Raspberry Pi 4	1
HC-05 Bluetooth Module	1
Flame Sensor	1
Ultrasonic Sensor HC-SR04	1
BTS7960 / IBT-2 Motor Driver	2
DC Gear Motors	2
2-Channel Relay Module	1
Mini Water Pump	1
Servo Motor (SG90)	1
Active Buzzer	1
LiPo Battery 1650mAh 3S	1
Tank Chassis Kit	1
Jumper Wires	Multiple

All hardware components were carefully selected to balance cost, performance, and availability in the local market.

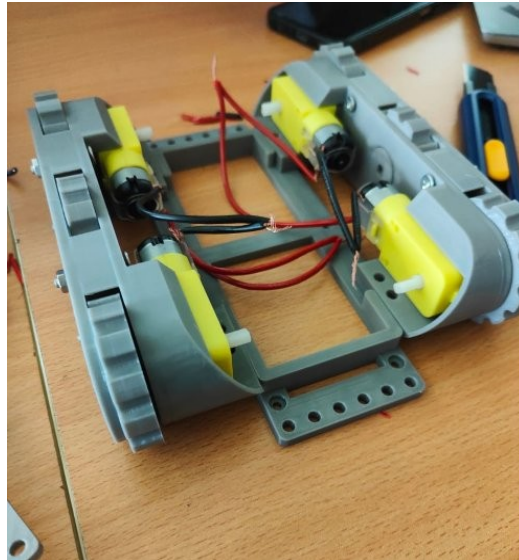


Figure 3: Robot Chassis with DC Motors

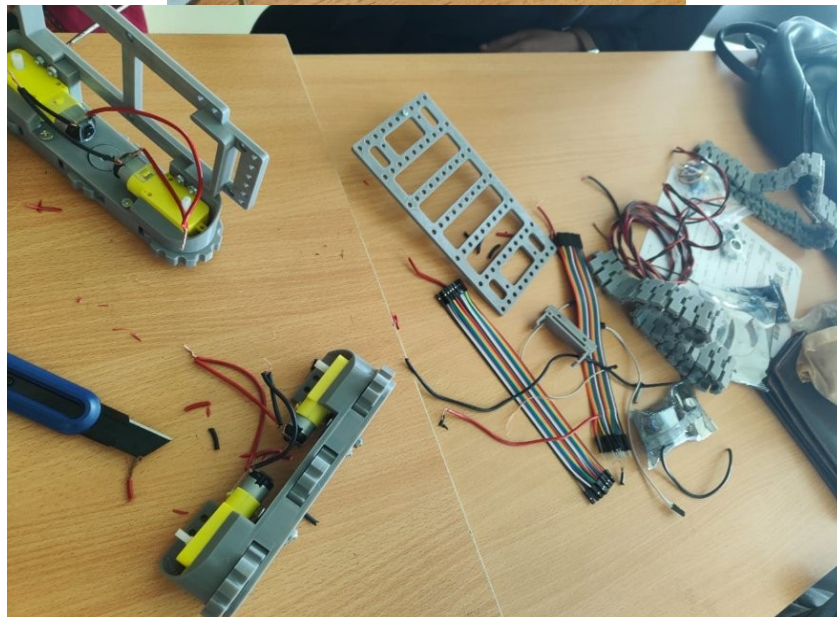


Figure 4: Components Layout on Table

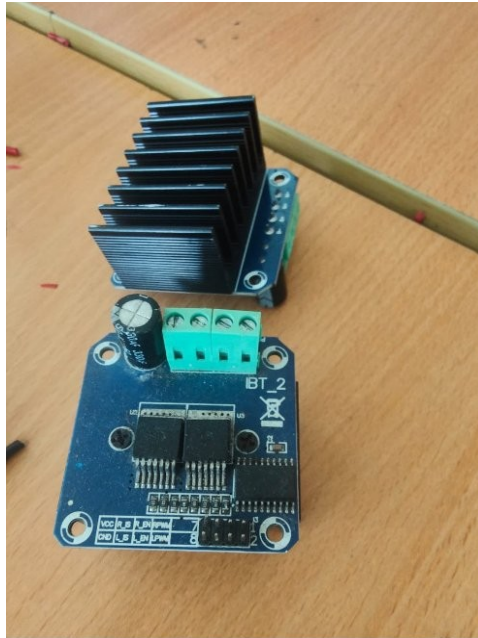


Figure 5: IBT-2 Motor Driver Modules

Figure 6: LiPo Battery 1650mAh 3S 30C



Figure 7: SG90 Servo Motor



Figure 8: Mini Water Pump with Tube

7. Software Tools

The following software tools were used during development:

- Arduino IDE — For writing and uploading Arduino firmware
- Raspberry Pi OS — Operating system for the Raspberry Pi 4
- Python 3 — Primary programming language for AI and vision scripts
- OpenCV Library — For image processing and camera input
- YOLO (You Only Look Once) — Real-time human detection model
- Bluetooth RC Controller App — Mobile app for robot control
- GitHub — Version control and project collaboration

8. Circuit Connections

HC-05 Bluetooth Module

- TX → Arduino D2
- RX → Arduino D3
- VCC → 5V | GND → GND

Flame Sensor

- OUT → Arduino D9
- VCC → 5V | GND → GND

Ultrasonic Sensor (HC-SR04)

- TRIG → A4 | ECHO → A5
- VCC → 5V | GND → GND

Relay Module

- IN → Arduino D4
- VCC → 5V | GND → GND

Left Motor Driver (BTS7960)

- RPWM → A0 | LPWM → A1 | REN → A2 | LEN → A3

Right Motor Driver (BTS7960)

- RPWM → D5 | LPWM → D6 | REN → D7 | LEN → D8

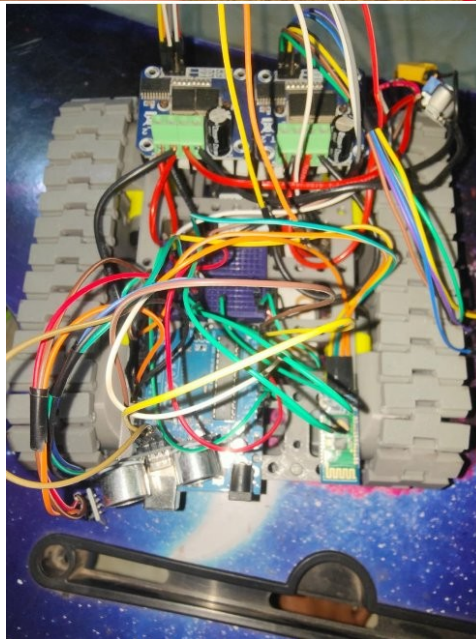
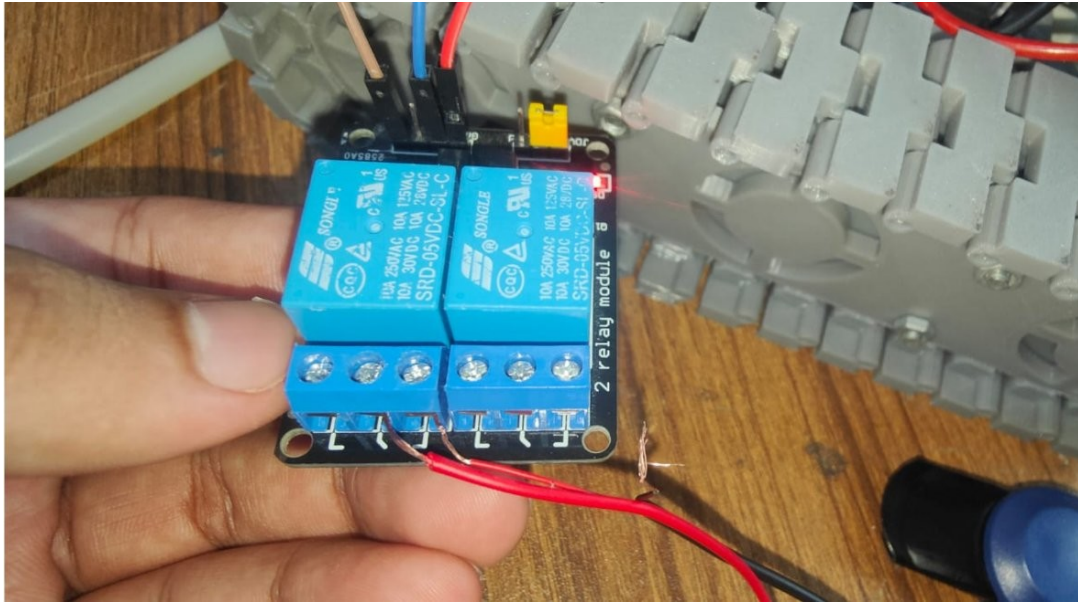


Figure 9: 2-Channel Relay Module

Figure 10: Full Wiring Inside Robot

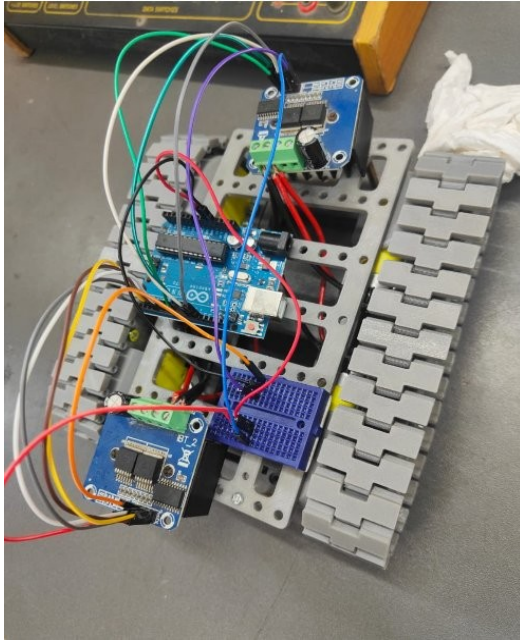


Figure 11: Arduino + Bluetooth Module Setup

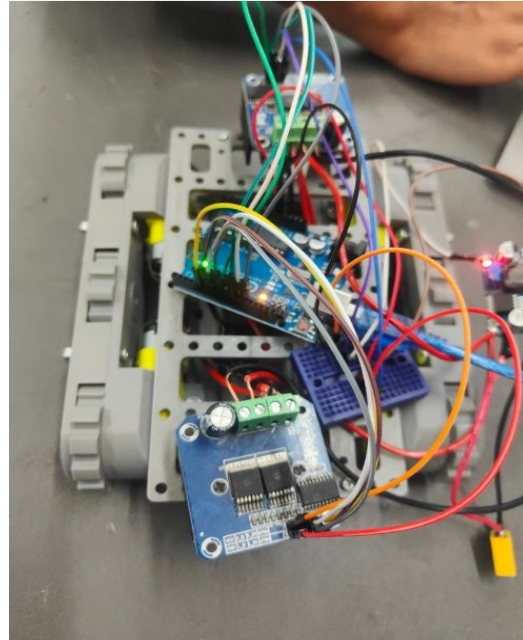


Figure 12: Robot Side View with Wiring

9. Robot Assembly Process

The robot was assembled in stages, starting from the chassis and motors to the electronics and final integration.

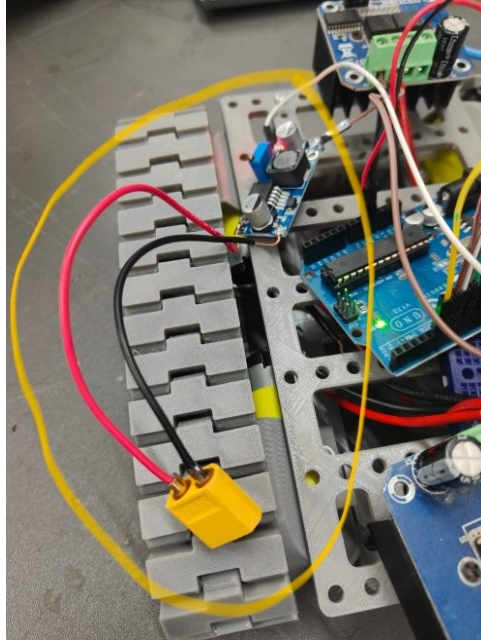


Figure 13: Battery Connector Close-up

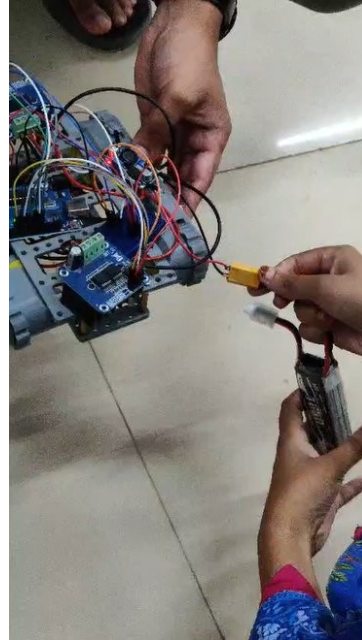


Figure 14: LiPo Battery Connection

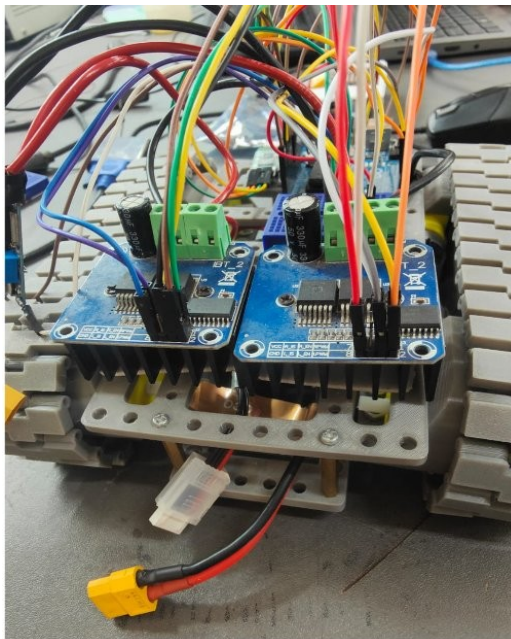


Figure 15: Dual IBT-2 Motor Drivers Mounted

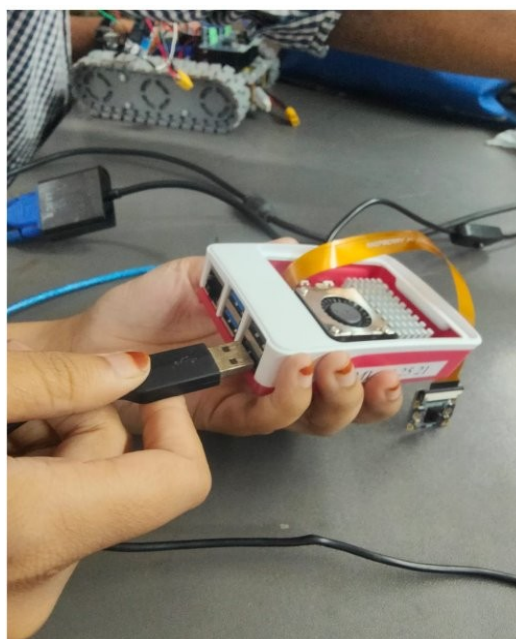


Figure 16: Raspberry Pi 4 Module

10. Working Principle

10.1 Bluetooth Control

The robot receives movement commands from a smartphone via the HC-05 Bluetooth module. Commands are processed by Arduino and translated into motor control signals.

- F = Forward | B = Backward | L = Left | R = Right | S = Stop

10.2 Fire Detection & Extinguishing

The flame sensor continuously monitors the environment for infrared radiation emitted by fire. When fire is detected:

- Active buzzer sounds an alert
- Relay module turns ON
- Water pump activates automatically
- Servo motor directs nozzle toward the fire source

10.3 Obstacle Detection

The ultrasonic sensor (HC-SR04) measures distance to nearby objects using sound waves. When an obstacle is detected within a safe threshold, the buzzer sounds and the operator is alerted.

10.4 Human Detection

The Raspberry Pi camera continuously captures video frames. Using the YOLO object detection model running on the Pi, human presence is detected in real-time. When a human is detected:

- Alert message is sent to Arduino via serial communication
- Robot stops automatically for safety
- "Human Detected! Robot Stopped." is logged in terminal



The screenshot shows a mobile application interface with a blue header bar labeled "Terminal". Below the header is a scrollable list of log entries. Each entry consists of a timestamp followed by a status message. The messages are color-coded: green for fire detection and pump status, and white for human detection and robot control status. At the bottom of the screen, there are six buttons labeled M1 through M6, and a large right-pointing arrow button.

```
3:29:21.043 FIRE DETECTED! PUMP ON
3:29:21.175 FIRE DETECTED! PUMP ON
3:29:21.224 FIRE DETECTED! PUMP ON
3:29:21.333 FIRE DETECTED! PUMP ON
3:29:21.394 FIRE DETECTED! PUMP ON
3:29:21.468 FIRE DETECTED! PUMP ON
3:29:21.523 FIRE DETECTED! PUMP ON
3:29:21.634 FIRE DETECTED! PUMP ON
3:29:21.664 FIRE DETECTED! PUMP ON
3:29:21.759 FIRE DETECTED! PUMP ON
3:29:21.815 FIRE DETECTED! PUMP ON
3:29:21.933 FIRE DETECTED! PUMP ON
3:29:21.959 FIRE DETECTED! PUMP ON
3:29:22.056 FIRE DETECTED! PUMP ON
3:29:22.119 FIRE DETECTED! PUMP ON
3:29:22.242 FIRE DETECTED! PUMP ON
3:29:22.261 FIRE DETECTED! PUMP ON
3:29:22.360 FIRE DETECTED! PUMP ON
3:29:22.414 FIRE DETECTED! PUMP ON
3:29:22.592 FIRE DETECTED! PUMP ON
3:29:22.592 Pi sent: H
3:29:23.069 Human Detected! Robot Stopped.
3:29:23.115 FIRE DETECTED! PUMP ON
3:29:23.168 Pi sent: N
3:29:23.236 No Human. Robot Control Active.
3:29:23.236 FIRE DETECTED! PUMP ON
3:29:23.333 Pi sent: H
3:29:23.835 Human Detected! Robot Stopped.
3:29:23.884 FIRE DETECTED! PUMP ON
3:29:23.923 FIRE DETECTED! PUMP ON
3:29:24.041 FIRE DETECTED! PUMP ON
3:29:24.103 FIRE DETECTED! PUMP ON
3:29:24.178 FIRE DETECTED! PUMP ON
3:29:24.233 FIRE DETECTED! PUMP ON
3:29:24.342 FIRE DETECTED! PUMP ON
3:29:24.404 FIRE DETECTED! P
```

Figure 17: Serial Terminal Output Showing Fire Detection and Human Detection Events

11. Results and Testing

Several tests were conducted to verify the performance of the robot system. All modules performed as expected during testing:

Test Case	Description	Result
Bluetooth Control	All movement commands (F/B/L/R/S) from mobile app	Pass ✓
Fire Detection	Flame sensor detected fire and activated relay + pump	Pass ✓
Obstacle Detection	Ultrasonic sensor detected nearby obstacles	Pass ✓
Human Detection	Raspberry Pi identified human presence via camera	Pass ✓
Water Pump Activation	Relay controlled water pump during fire detection	Pass ✓
Robot Chassis Movement	Tank-track movement on flat surfaces verified	Pass ✓
Servo Motor Control	Servo rotated water nozzle direction successfully	Pass ✓

The serial monitor output (Figure 17) confirmed successful fire detection events ("FIRE DETECTED! PUMP ON") and human detection events ("Human Detected! Robot Stopped."), validating the integration between Raspberry Pi and Arduino.

12. Real-Life Applications

This project can be deployed in a wide range of real-world scenarios:

- Fire rescue operations in buildings and industrial facilities
- Industrial safety monitoring in factories and warehouses
- Chemical plants and hazardous material storage areas
- Smart safety systems in public infrastructure
- Emergency first response operations
- Search and rescue operations in disaster zones

The system improves safety by reducing direct human exposure to dangerous fire environments.

13. Advantages

- Low-cost implementation using widely available components
- Easy wireless operation via Bluetooth mobile app
- Automatic fire extinguishing without human intervention
- Real-time human detection using AI (YOLO)
- Tank-track chassis for improved mobility on uneven surfaces
- Servo-controlled nozzle for directional fire suppression
- Obstacle detection for safe autonomous navigation

14. Limitations

- Limited battery backup from LiPo 1650mAh battery
- Human detection performance decreases in low-light conditions
- Suitable for small-to-medium scale fire incidents
- Bluetooth range limited to approximately 10 meters
- Requires flat or semi-flat terrain for optimal movement

15. Cost Analysis

The following cost breakdown represents the total hardware investment for the project:

Component	Cost (BDT)
Arduino Uno	700
Raspberry Pi 4	9000
DC Motors & Chassis	800
BTS7960 Motor Drivers (x2)	1000
Sensors (Flame, Ultrasonic)	700
Relay Module & Water Pump	600
LiPo Battery 1650mAh	1200
Other Components	1000
Total Cost	≈ 15,000

The project is highly cost-effective compared to commercial rescue robots, which typically cost several hundred thousand BDT, while still providing multiple intelligent safety features.

16. Future Work

The following improvements are planned for future versions of the rescue robot:

- GPS Tracking System for precise location sharing
- IoT-based Monitoring via cloud dashboard
- Dedicated Mobile Application with live video feed
- Autonomous Navigation using machine learning
- Cloud Data Logging for incident records
- Live Video Streaming from Raspberry Pi camera
- Advanced AI Object Detection with thermal imaging
- Real-Time Location Sharing with rescue teams

17. Conclusion

This project successfully developed an AI-Based Fire Fighting and Human Detection Rescue Robot capable of detecting fire, obstacles, and human presence. The integration of Arduino Uno and Raspberry Pi provides intelligent monitoring, wireless control, and automated fire-fighting functionality.

The project demonstrates the practical application of Artificial Intelligence, Embedded Systems, and Robotics in improving safety and rescue operations. The tank-based chassis provides reliable movement, while the combination of sensors and AI enables real-time decision-making without human intervention.

Future enhancements such as autonomous navigation, GPS tracking, and IoT connectivity can further improve system autonomy, reliability, and real-world applicability — making it a valuable tool for emergency response teams.

References

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2. Raspberry Pi Official Documentation — <https://www.raspberrypi.com/documentation/>
3. OpenCV Documentation — <https://docs.opencv.org/>
4. YOLO Object Detection — <https://github.com/ultralytics/yolov5>
5. HC-SR04 Ultrasonic Sensor Datasheet
6. HC-05 Bluetooth Module Datasheet
7. BTS7960 Motor Driver Datasheet
8. GitHub Repository — <https://github.com/shafinSI/AI-Fire-Fighting-Human-Detection-Rescue-Robot>